

Obstacle Avoiding Car (4 Motors)

Components

- Arduino UNO
- HC-SR04 Ultrasonic Sensor
- L298N Motor Driver
- 4 DC Gear Motors (2 Left + 2 Right)
- Robot Chassis
- Battery
- Jumper Wires

Wiring

HC-SR04

VCC → 5V

GND → GND

TRIG → D9

ECHO → D10

L298N

ENA → D5

IN1 → D2

IN2 → D3

IN3 → D4

IN4 → D7

ENB → D6

Left Front + Left Rear → OUT1 & OUT2

Right Front + Right Rear → OUT3 & OUT4

Complete Arduino Code

```
#define TRIG 9
#define ECHO 10
#define ENA 5
#define IN1 2
#define IN2 3
#define IN3 4
#define IN4 7
#define ENB 6

long duration;
int distance;

void setup() {
  pinMode(TRIG,OUTPUT);
  pinMode(ECHO,INPUT);
  pinMode(ENA,OUTPUT);
  pinMode(ENB,OUTPUT);
  pinMode(IN1,OUTPUT);
  pinMode(IN2,OUTPUT);
  pinMode(IN3,OUTPUT);
  pinMode(IN4,OUTPUT);
}

int getDistance(){
  digitalWrite(TRIG,LOW); delayMicroseconds(2);
  digitalWrite(TRIG,HIGH); delayMicroseconds(10);
  digitalWrite(TRIG,LOW);
  duration=pulseIn(ECHO,HIGH);
  return duration*0.034/2;
}

void forward(){
```

```
    analogWrite(ENA,180); analogWrite(ENB,180);
    digitalWrite(IN1,HIGH); digitalWrite(IN2,LOW);
    digitalWrite(IN3,HIGH); digitalWrite(IN4,LOW);
}

void backward(){
    digitalWrite(IN1,LOW); digitalWrite(IN2,HIGH);
    digitalWrite(IN3,LOW); digitalWrite(IN4,HIGH);
}

void rightTurn(){
    digitalWrite(IN1,HIGH); digitalWrite(IN2,LOW);
    digitalWrite(IN3,LOW); digitalWrite(IN4,HIGH);
}

void stopCar(){
    digitalWrite(IN1,LOW); digitalWrite(IN2,LOW);
    digitalWrite(IN3,LOW); digitalWrite(IN4,LOW);
}

void loop(){
    distance=getDistance();
    if(distance>20) forward();
    else{
        stopCar(); delay(200);
        backward(); delay(400);
        rightTurn(); delay(500);
    }
}
```